

HOMEWORK 13

Due date: Monday of Week 14

Exercises: 12.4, M.1, M.2, M.9, M.10, M.14. pages 75-77.

Exercises: 7.7, 7.8, 7.9, 7.10, 8.1, 8.2, 8.4, 11.1, 11.2, 11.3, 11.5, 11.8, M.7, pages 191-194.

Exercise: 5.7, page 223.

1. COSETS, DOUBLE COSETS, SEMI-DIRECT PRODUCT

Problem 1. Let G be a group (not necessarily finite) and $H < G$ be a subgroup of G . Recall that G/H denotes the set of all left H -cosets and $H \backslash G$ denotes the set of all right H -cosets. Show that $f : G/H \rightarrow H \backslash G$ defined by $f(gH) = Hg^{-1}$ is well-defined and defines a bijection between G/H and $H \backslash G$. In particular, the number of left cosets is the same as the number of right cosets.

There are at least two elements ϕ_0, ϕ_1 in $\text{Aut}(\mathbb{Z}/n\mathbb{Z})$ defined by

$$\phi_0 = \text{id}_{\mathbb{Z}/n\mathbb{Z}}; \phi_1(x) = -x, \forall x \in \mathbb{Z}/n\mathbb{Z}.$$

Consider the map

$$\phi : \mathbb{Z}/2\mathbb{Z} \rightarrow \text{Aut}(\mathbb{Z}/n\mathbb{Z})$$

defined by $\phi(\bar{0}) = \phi_0, \phi(\bar{1}) = \phi_1$. It is clear that ϕ is a group homomorphism.

Problem 2. Show that $\mathbb{Z}/n\mathbb{Z} \rtimes_{\phi} \mathbb{Z}/2\mathbb{Z}$ is isomorphic to D_n , the dihedral group of order $2n$.

The next several problems are about double cosets, and most of them could be in last HW.

Problem 3. Let F be a field and let $B_n(F) \subset \text{GL}_n(F)$ be the upper triangular subgroup.

- (1) Determine the double cosets $B_2(F) \backslash \text{GL}_2(F) / B_2(F)$.
- (2) How about $B_n(F) \backslash \text{GL}_n(F) / B_n(F)$?

This problem might be hard. It is related to the *UPL* (upper triangular, permutation subgroup, and lower triangular subgroup) decomposition of a matrix, See HW 3, Problem 5 of last year. If you don't know how to do the general problem, try the case when $n = 2$ and $F = \mathbb{F}_2$ (or $\mathbb{F}_3$).

Let $G \times X \rightarrow X$ be an action of a group G on a set X . Recall that $G \backslash X$ denote the set of orbits.

Problem 4. Let G be a group and H, K are subgroups of G . Show the following basic properties of double cosets.

- (1) For $x \in G$, the double coset HxK is a union of right H -cosets and a union of left K -cosets. More precisely,

$$HxK = \coprod_{Hxk \in H \backslash HxK} Hxk = \coprod_{hxK \in HxK/K} hxK.$$

- (2) Let G act on the left cosets G/K from the left by $x.(gK) = (xg)K$. See Section 6.8 of Artin. We restrict this action to H and consider the action

$$H \times G/K \rightarrow G/K$$

defined by $(h, gK) = (hg)K$. Show that there is a bijection between the double coset $H \backslash G/K$ and the set of orbits $H \backslash (G/K)$. This explains that the notation is consistent. There is a similar statement when we switch the role of H and K .

- (3) Suppose that all groups are finite. For $x \in G$, show that

$$|HxK| = [H : H \cap xKx^{-1}]|K| = [K : K \cap x^{-1}Hx]|H|.$$

(4) Show that

$$[G : H] = \sum_{HxK \in H \backslash G / K} [K : K \cap x^{-1}Hx]$$

and

$$[G : K] = \sum_{HxK \in H \backslash G / K} [H : H \cap xKx^{-1}].$$

(5) Consider the group action of $(H \times K)$ on G defined by

$$((h, k), g) = h g k^{-1}, (h, k) \in H \times K, g \in G.$$

Check that this is a group action and there is a bijection between $H \backslash G / K$ and the orbits of this action.

(6) Suppose G is finite. For $h \in H, k \in K$, consider $G^{(h,k)} = \{g \in G : h g k = g\}$. Show that

$$|H \backslash G / K| = \frac{1}{|H||K|} \sum_{(h,k) \in H \times K} |G^{(h,k)}|.$$

For the last one, use Ex. M.7, page 194 of Artin. The other parts are routine.

2. EXAMPLES OF GROUP ACTIONS, REGULAR POLYHEDRAL

Problem 5. Determine the order of the group $\mathrm{GL}_n(\mathbb{F}_p)$ and $\mathrm{Sp}_{2n}(\mathbb{F}_p)$, where p is a prime number with $p \neq 2$.

Hint: Consider the action of $\mathrm{GL}_n(\mathbb{F}_p)$ on $\mathbb{F}_p^n$ by left multiplication. Recall that $\mathrm{Sp}_{2n}(F)$ is the group $G = \{g \in \mathrm{GL}_{2n}(F) \mid g^t J_{2n} g = J_{2n}\}$, for a field F of odd characteristic. Here J_{2n} is any non-degenerate anti-symmetric matrix of rank $2n$. One can take

$$J_{2n} = \begin{pmatrix} & I_n \\ -I_n & \end{pmatrix},$$

or

$$J_{2n} = \begin{pmatrix} & & & 1 \\ & & 1 & \\ & & \cdots & \\ & -1 & & \\ -1 & & & \end{pmatrix},$$

where the number of $+1$ is n in the anti-diagonal.

Problem 6. Let $G = D_4$, which is the symmetric group of a square. Let X be the set of two diagonals of this square. Analyze the action of G on X and describe the map $D_4 \rightarrow \mathrm{Perm}(X) \cong S_2$ obtained from this action.

Problem 7. Let T be a tetrahedron and let $G = G_T$ be the symmetric group of T .

- (1) Let V be the set of vertices of T and consider the action $G \times V \rightarrow V$. This will give a map $G \rightarrow \mathrm{Perm}(V) \cong S_4$. From this to deduce that $G \cong A_4$.
- (2) Let X be the set of the 3 line segments which connect the middle points of the opposite edges. Consider the action $G \times X \rightarrow X$. Consider the homomorphism $G \rightarrow \mathrm{Perm}(X) \cong S_3$ obtained from this action. Show that this homomorphism is just composition $A_4 \hookrightarrow S_4 \rightarrow S_3$ (up to a relabeling of indices), where the map $S_4 \rightarrow S_3$ is the one in Example 2.5.13, page 50-51 of Artin's book.

For part (1), use Exercise 5.7 of page 223.

Let $X_n = \{x_1, \dots, x_n\}$ be a set of n elements. Then S_n is just $\mathrm{Perm}(X_n)$. For a subgroup $G \subset S_n$, we can consider the natural action of G on X_n . A subgroup of G is called **transitive**, if it acts transitively on X_n . Let $G = \{g \in S_n \mid g x_n = x_n\}$. Note that $G \cong S_{n-1}$, but G is not a transitive subgroup of S_n .

Problem 8. Let C be a cube and let $G = G_C$ be the symmetric group of C .

- (1) Let X be the set of the 4 diagonals of C . A diagonal of C is just a line segment which connects 2 opposite vertices. Consider the action $G \rightarrow X \rightarrow X$. Show that $G \cong S_4$.
- (2) Using the first part to classify all transitive subgroups of S_4 .
- (3) Let F be the set of the 6 faces of C . Consider the action of G on F and the homomorphism $G \rightarrow \text{Perm}(F) = S_6$. Is this map injective? Is the image a transitive subgroup of S_6 ?
- (4) Let Y be the set of the 3 pairs of opposite faces. Then $|Y| = 3$. Show that G acts on Y and the action gives the homomorphism $S_4 \rightarrow S_3$ in Example 2.5.13, page 50-51 of Artin's book.
- (5) Call the 8 vertices as $v_1, \dots, v_8$ such that v_1, v_2, v_3, v_4 is the top square and v_5, v_6, v_7, v_8 is the bottom square and v_i is above v_{i+4} . (The picture is the same as what I drew in class. I will try to provide a picture if I have more time). Let T be the tetrahedron with vertices v_1, v_3, v_6, v_8 . Describe the orbit and stabilizer of T under the natural action.

For an answer of (2), see (16.9.1), page 493 of Artin's book. We will use this result later.

Problem 9. Let I be an icosahedron and let $G = G_I$ be the symmetric group of I . Then I has 12 vertices. Let X be the set of 6 main diagonals. Show that G acts on X and this gives a homomorphism $f : G \rightarrow \text{Perm}(X) = S_6$. Show that f is injective and its image is a transitive subgroup of S_6 .

We know that $G_I \cong A_5$ (this will be proved in next week, see page 199-200 of Artin's book) and thus S_6 contains a transitive subgroup which is isomorphic to A_5 . On the contrary, if we view A_5 as a subgroup of S_6 by identifying it with as a subgroup of $H = \{h \in S_6 \mid hx_6 = x_6\}$, say, then it is not transitive. Recall that we know $\text{Aut}(S_6)$ is strictly larger than S_6 . In fact, we have $\text{Out}(S_6) = C_2$. Thus there is a unique outer automorphism of S_6 (up to inner automorphisms). This unique outer automorphism is related to this transitive subgroup A_5 of S_6 as in Problem 9. The construction of such an outer automorphism is given in next homework.

We next give a different way to construct a transitive subgroup of S_6 , which is isomorphic to A_5 .

3. PROJECTIVE SPACE

Let F be a field and n be a positive integer. We introduce an equivalence relation on $F^{n+1} - \{0\}$ as follows. For $\alpha, \beta \in F^{n+1} - \{0\}$, we say that $\alpha \sim \beta$ if there is an element $c \in F^\times$ such that $\beta = c\alpha$.

Problem 10. Show that $R = \{(\alpha, \beta) \in (F^{n+1} - \{0\}) \times (F^{n+1} - \{0\}) : \alpha \sim \beta\}$ defines an equivalence relation on $F^{n+1} - \{0\}$.

Denote $\mathbb{P}^n(F) := (F^{n+1} - \{0\}) / \sim$, which is the set of equivalence classes of the above equivalence relation. For example, $\mathbb{P}^1(F)$ is just the set of all lines which passes the origin. The set $\mathbb{P}^n(F)$ is called the n -dimensional projective space. Here we only view it as a set even it has more structures.

An element of $\mathbb{P}^n(F)$ is of the form $[\alpha]$, where $\alpha \in F^{n+1} - \{0\}$ and $[\alpha]$ denotes the equivalence class of α . Note that $[\alpha] = [\beta]$ if and only if $\beta = c\alpha$ for some $c \in F^\times$.

There is a group action of $\text{GL}_{n+1}(F)$ on $\mathbb{P}^n(F)$ defined by

$$g.[\alpha] = [g\alpha], g \in \text{GL}_{n+1}(F), \alpha \in F^{n+1}.$$

Consider the group $\text{PGL}_{n+1}(F) = \text{GL}_{n+1}(F)/Z$, where $Z = \{aI_{n+1}, a \in F^\times\}$ is the center of $\text{GL}_{n+1}(F)$. The group $\text{PGL}_{n+1}(F)$ is called the **projective general linear group** over F . An element of $\text{PGL}_{n+1}(F)$ is of the form gZ , where $g \in \text{GL}_{n+1}(F)$ and $gZ = hZ$ if and only if $g^{-1}h \in Z$, or $g = hz$ for some $z \in Z$. Similarly, we consider the group $\text{PSL}_{n+1}(F) = \text{SL}_{n+1}(F)/Z_0$, where $Z_0 = \{aI_{n+1} : a \in F^\times, a^{n+1} = 1\}$ is the center of $\text{SL}_{n+1}(F)$.

Problem 11. Show that the map $\text{PGL}_{n+1}(F) \times \mathbb{P}^n(F) \rightarrow \mathbb{P}^n(F)$ defined by

$$(gZ, [\alpha]) \mapsto [g\alpha]$$

is well-defined and it defines a group action of $\text{PGL}_{n+1}(F)$ on $\mathbb{P}^n(F)$. Similarly, show that there is a group action of $\text{PSL}_{n+1}(F)$ on $\mathbb{P}^n(F)$.

We next consider some very special cases.

Problem 12. Let $\mathbb{F}$ be a finite field of order q . What is the cardinality of $\mathbb{P}^n(\mathbb{F})$? What is the cardinality of $\text{PGL}_n(\mathbb{F})$? What is the cardinality of $\text{PSL}_2(\mathbb{F})$?

Here “cardinality” of a finite set means the number of elements in the finite set. For the second part and third part, think about how many elements are in the group $\mathrm{GL}_n(\mathbb{F})$ and $\mathrm{SL}_2(\mathbb{F})$.

Problem 13. Let $\mathbb{F}_2$ be a finite field of 2 elements. Consider the action of $\mathrm{PSL}_2(\mathbb{F}_2)$ on $\mathbb{P}^1(\mathbb{F}_2)$. Show that

$$\mathrm{PSL}_2(\mathbb{F}_2) \cong S_3.$$

Similarly, show that

$$\mathrm{PSL}_2(\mathbb{F}_3) \cong A_4.$$

Problem 14 (No need to submit a solution). Let $\mathbb{F}_4$ be a finite field of 4 elements. Consider the action of $\mathrm{PSL}_2(\mathbb{F}_4)$ on $\mathbb{P}^1(\mathbb{F}_4)$. Show that

$$\mathrm{PGL}_2(\mathbb{F}_4) \cong A_5.$$

There is indeed a finite field of 4 elements. See HW 2 of last year. The above two problems are indeed Exercises 8.2, 8.3, page 287 of Artin’s book. Hint: The following fact will be useful. If H is a subgroup of S_n of index 2, then $H = A_n$. This Exercise 5.7, page 223.

Problem 15 (No need to submit a solution). Consider the action of $\mathrm{PSL}_2(\mathbb{F}_5)$ on $\mathbb{P}^1(\mathbb{F}_5)$ and the homomorphism $\mathrm{PSL}_2(\mathbb{F}_5) \rightarrow S_6 = \mathrm{Perm}(\mathbb{P}^1(\mathbb{F}_5))$ obtained from that. Show that this map is injective and its image is a transitive subgroup of S_6 .

One can show that $\mathrm{PSL}_2(\mathbb{F}_5) \cong A_5$ and the embedding $A_5 \rightarrow S_6$ obtained from the above map is roughly the same as the embedding one obtained from Problem 9. Here “roughly” means, one need to choose the indices carefully.